

Table 1. Pin Description

Symbol	Type	Name and Function																																								
A ₈ -A ₁₅	O	ADDRESS BUS: The most significant 8 bits of memory address or the 8 bits of the I/O address, 3-stated during Hold and Halt modes and during RESET.																																								
AD ₀₋₇	I/O	MULTIPLEXED ADDRESS/DATA BUS: Lower 8 bits of the memory address (or I/O address) appear on the bus during the first clock cycle (T state) of a machine cycle. It then becomes the data bus during the second and third clock cycles.																																								
ALE	O	ADDRESS LATCH ENABLE: It occurs during the first clock state of a machine cycle and enables the address to get latched into the on-chip latch of peripherals. The falling edge of ALE is set to guarantee setup and hold times for the address information. The falling edge of ALE can also be used to strobe the status information. ALE is never 3-stated.																																								
S ₀ , S ₁ and IO/ $\overline{M}$	O	MACHINE CYCLE STATUS: <table><tr><th>IO/$\overline{M}$</th><th>S₁</th><th>S₀</th><th>Status</th></tr><tr><td>0</td><td>0</td><td>1</td><td>Memory write</td></tr><tr><td>0</td><td>1</td><td>0</td><td>Memory read</td></tr><tr><td>1</td><td>0</td><td>1</td><td>I/O write</td></tr><tr><td>1</td><td>1</td><td>0</td><td>I/O read</td></tr><tr><td>0</td><td>1</td><td>1</td><td>Opcode fetch</td></tr><tr><td>1</td><td>1</td><td>1</td><td>Interrupt Acknowledge</td></tr><tr><td>*</td><td>0</td><td>0</td><td>Halt</td></tr><tr><td>*</td><td>X</td><td>X</td><td>Hold</td></tr><tr><td>*</td><td>X</td><td>X</td><td>Reset</td></tr></table> * = 3-state (high impedance) X = unspecified S₁ can be used as an advanced R/$\overline{W}$ status. IO/$\overline{M}$, S₀ and S₁ become valid at the beginning of a machine cycle and remain stable throughout the cycle. The falling edge of ALE may be used to latch the state of these lines.	IO/ $\overline{M}$	S ₁	S ₀	Status	0	0	1	Memory write	0	1	0	Memory read	1	0	1	I/O write	1	1	0	I/O read	0	1	1	Opcode fetch	1	1	1	Interrupt Acknowledge	*	0	0	Halt	*	X	X	Hold	*	X	X	Reset
IO/ $\overline{M}$	S ₁	S ₀	Status																																							
0	0	1	Memory write																																							
0	1	0	Memory read																																							
1	0	1	I/O write																																							
1	1	0	I/O read																																							
0	1	1	Opcode fetch																																							
1	1	1	Interrupt Acknowledge																																							
*	0	0	Halt																																							
*	X	X	Hold																																							
*	X	X	Reset																																							
$\overline{RD}$	O	READ CONTROL: A low level on $\overline{RD}$ indicates the selected memory or I/O device is to be read and that the Data Bus is available for the data transfer, 3-stated during Hold and Halt modes and during RESET.																																								
$\overline{WR}$	O	WRITE CONTROL: A low level on $\overline{WR}$ indicates the data on the Data Bus is to be written into the selected memory or I/O location. Data is set up at the trailing edge of $\overline{WR}$. 3-stated during Hold and Halt modes and during RESET.																																								
READY	I	READY: If READY is high during a read or write cycle, it indicates that the memory or peripheral is ready to send or receive data. If READY is low, the CPU will wait an integral number of clock cycles for READY to go high before completing the read or write cycle. READY must conform to specified setup and hold times.																																								
HOLD	I	HOLD: Indicates that another master is requesting the use of the address and data buses. The CPU, upon receiving the hold request, will relinquish the use of the bus as soon as the completion of the current bus transfer. Internal processing can continue. The processor can regain the bus only after the HOLD is removed. When the HOLD is acknowledged, the Address, Data $\overline{RD}$, $\overline{WR}$, and IO/ $\overline{M}$ lines are 3-stated.																																								
HLDA	O	HOLD ACKNOWLEDGE: Indicates that the CPU has received the HOLD request and that it will relinquish the bus in the next clock cycle. HLDA goes low after the Hold request is removed. The CPU takes the bus one half clock cycle after HLDA goes low.																																								
INTR	I	INTERRUPT REQUEST: Is used as a general purpose interrupt. It is sampled only during the next to the last clock cycle of an instruction and during Hold and Halt states. If it is active, the Program Counter (PC) will be inhibited from incrementing and an $\overline{INTA}$ will be issued. During this cycle a RESTART or CALL instruction can be inserted to jump to the interrupt service routine. The INTR is enabled and disabled by software. It is disabled by Reset and immediately after an interrupt is accepted.																																								